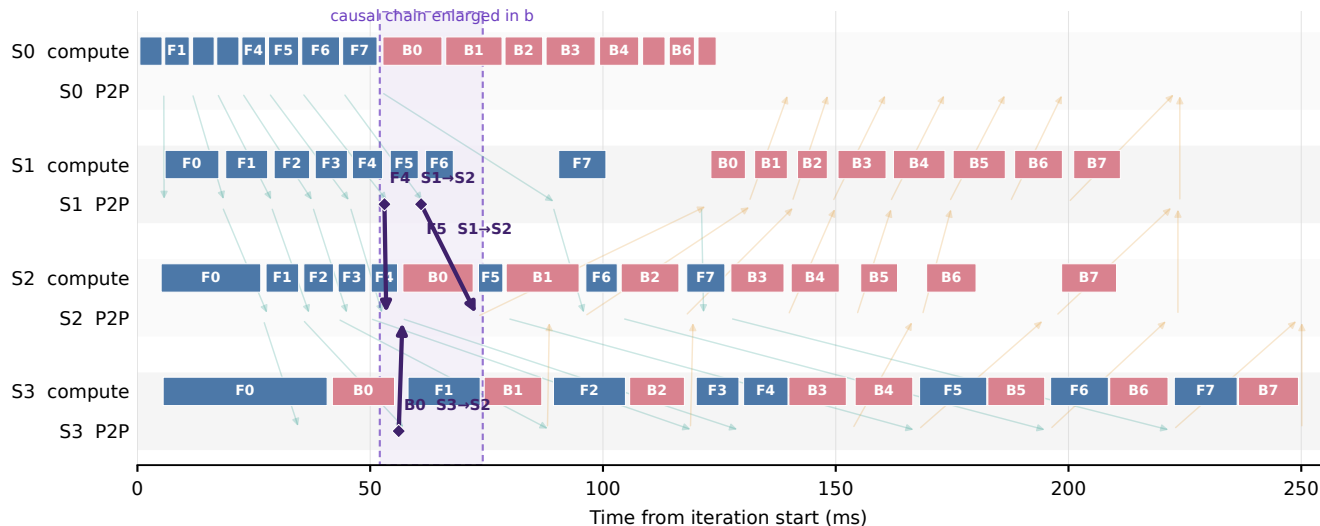
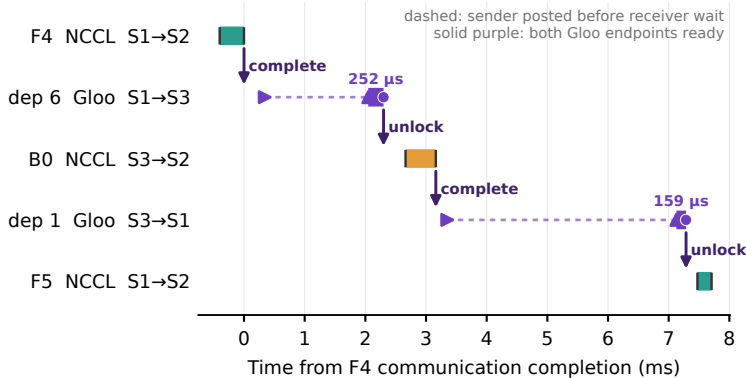


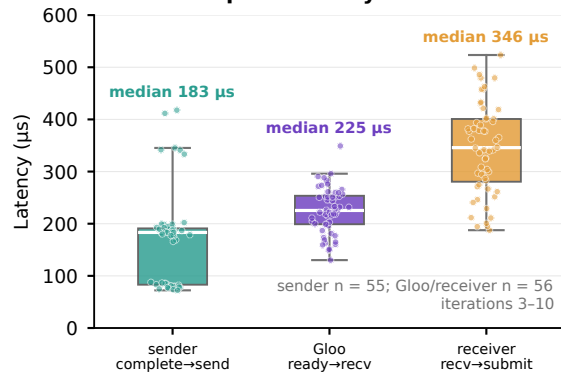
a C schedule: four-stage execution trace (iteration 5)



b Measured cross-rank dependency chain



c Control-path latency



Forward compute Backward compute Forward communication Backward communication Gloo control signal